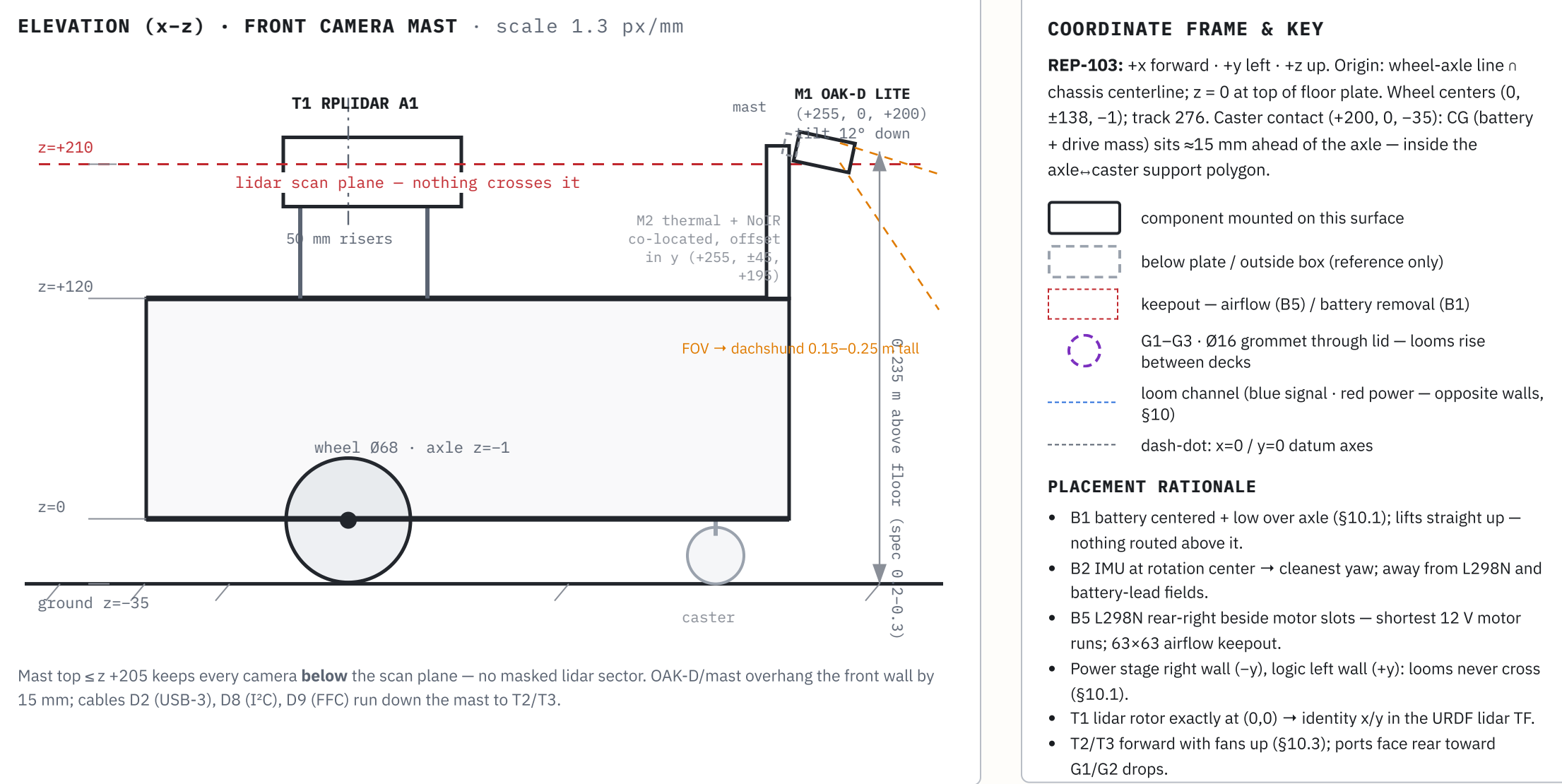
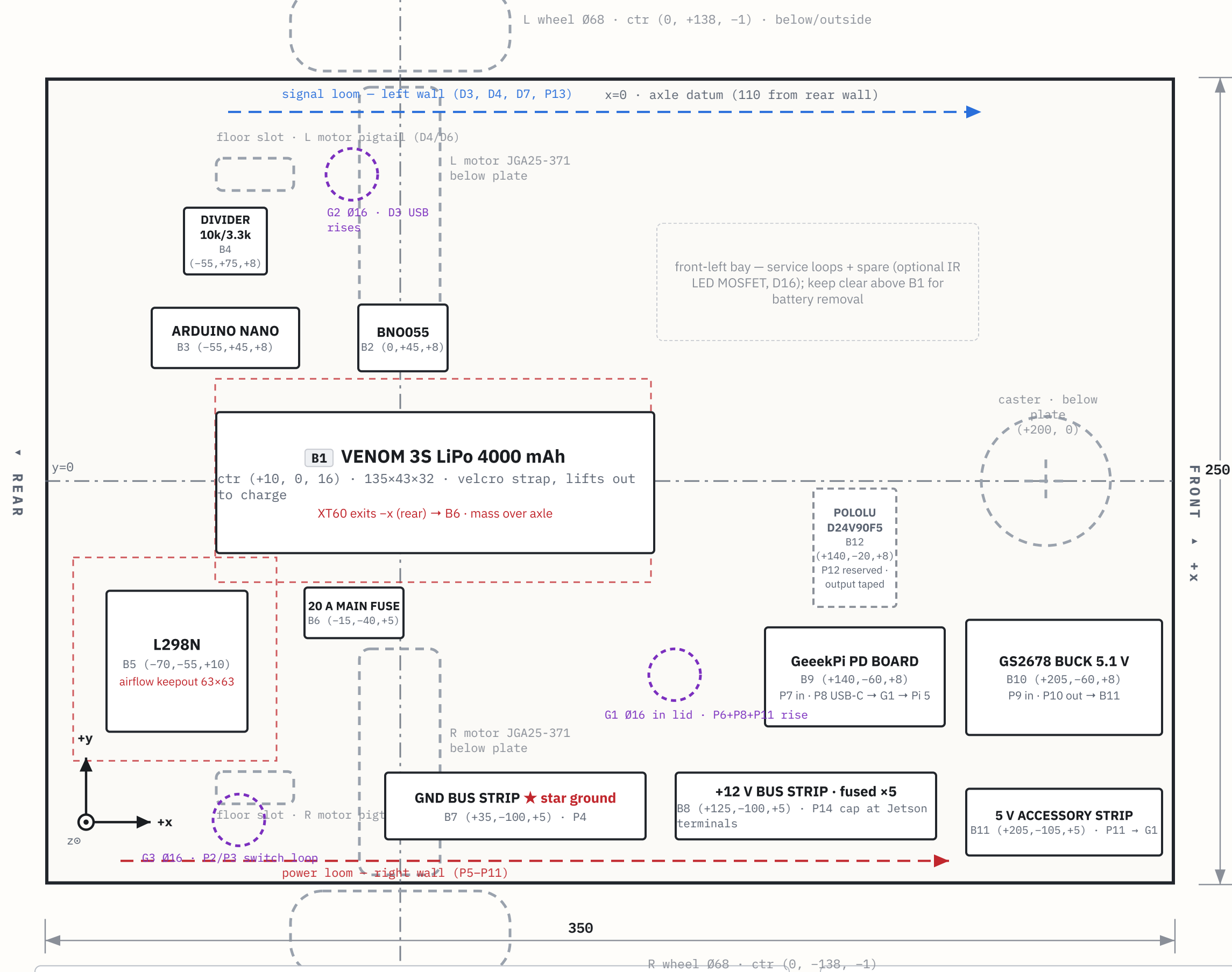


BILLIEBOT — PHYSICAL COMPONENT LAYOUT

Companion to Master Wiring Schematic (Sheet 1) · connection IDs P1-P14 / D1-D16 unchanged · no wiring shown on this sheet

LAYOUT 1 · LOWER DECK — inside surface of box floor · mounting plane z = 0



COORDINATE FRAME & KEY

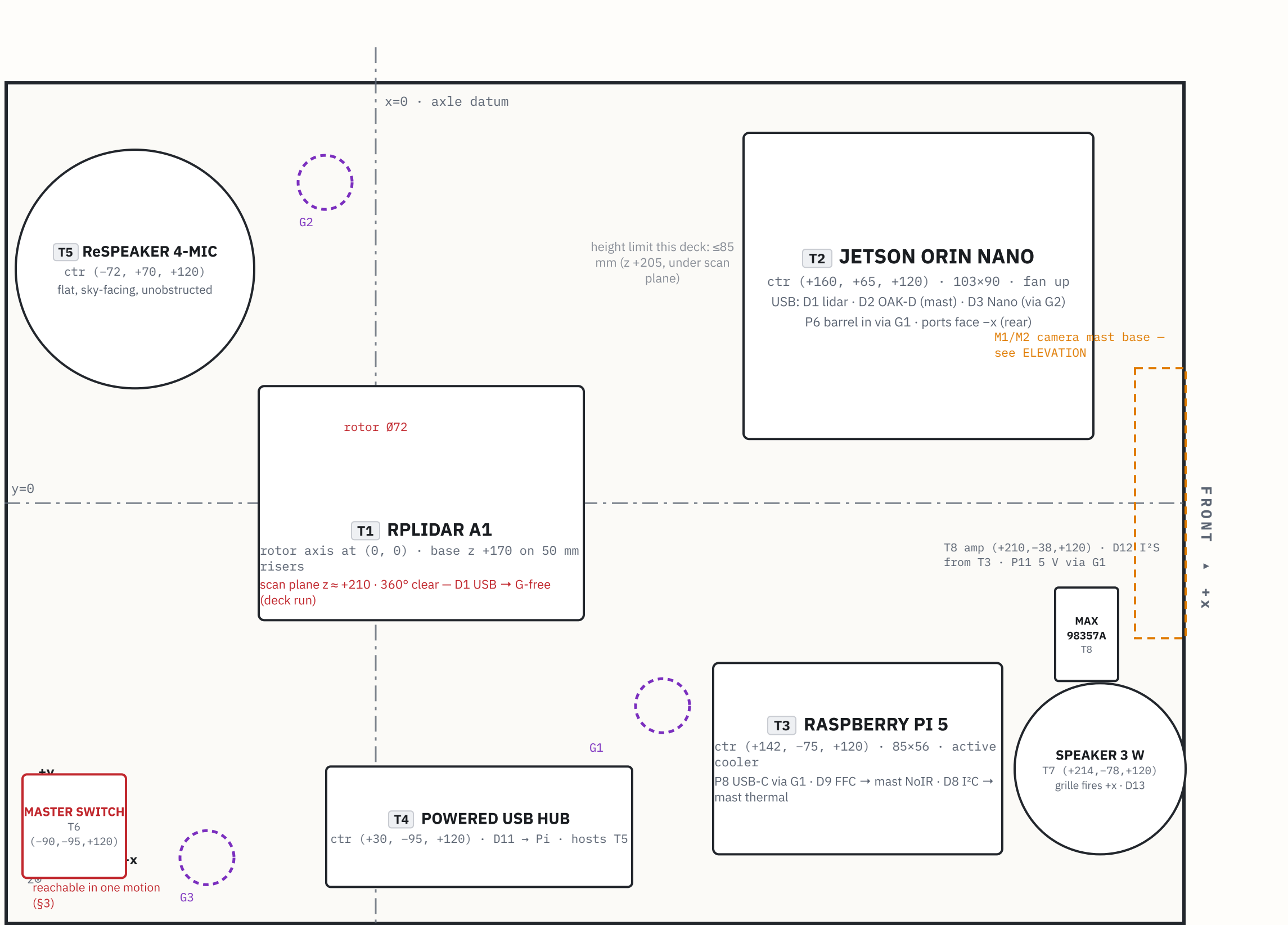
REP-103: +x forward · +y left · +z up. Origin: wheel-axle line n chassis centerline; z = 0 at top of floor plate. Wheel centers (0, ±138, -1); track 276. Caster contact (+200, 0, -35): CG (battery + drive mass) sits ≈15 mm ahead of the axle — inside the axle-caster support polygon.

- component mounted on this surface
- below plate / outside box (reference only)
- keepout — airflow (B5) / battery removal (B1)
- G1-G3 · Ø16 grommet through lid — looms rise between decks
- loom channel (blue signal · red power — opposite walls, §10)
- dash-dot: x=0 / y=0 datum axes

PLACEMENT RATIONALE

- B1 battery centered + low over axle (§10.1); lifts straight up — nothing routed above it.
- B2 IMU at rotation center → cleanest yaw; away from L298N and battery-lead fields.
- B5 L298N rear-right beside motor slots — shortest 12 V motor runs; 63×63 airflow keepout.
- Power stage right wall (-y), logic left wall (+y): looms never cross (§10.1).
- T1 lidar rotor exactly at (0,0) → identity x/y in the URDF lidar TF.
- T2/T3 forward with fans up (§10.3); ports face rear toward G1/G2 drops.
- T5 mic rear-left, away from speaker T7 (front-right) — minimizes echo pickup.
- T6 switch at rear corner: one-motion reach while robot faces

LAYOUT 2 · UPPER DECK — top surface of box lid · mounting plane z = +120



COMPONENT LOCATION TABLE · centers, mm, robot frame · transcribe to MEASURE_ME.md / URDF after build

LOWER DECK (z ref = 0)			
B1	Venom 3S LiPo 4000 mAh	(+10, 0, 0-32)	velcro strap; XT60 rear; over axle
B2	BNO055 IMU	(0, +45, +8)	at rotation center; D7 I²C to B3
B3	Arduino Nano V3	(-55, +45, +8)	USB up G2 → Jetson (D3)
B4	Battery-sense divider	(-55, +75, +8)	10k/3.3k; P13 to B3 A-pin
B5	L298N driver	(-70, -55, +10)	airflow keepout 63×63; P5, D5, D6
B6	20 A main fuse + XT60	(-15, -40, +5)	P1; battery lead reaches w/ slack
B7	GND bus strip	(+35, -100, +5)	★ single star-ground (P4)
B8	+12 V bus strip, 5 fuses	(+125, -100, +5)	P3; P14 cap at Jetson terminals
B9	GeekPi PD board	(+140, -60, +8)	P7 in; P8 USB-C up G1 → Pi
B10	GS2678 buck (5.1 V)	(+205, -60, +8)	P9 in; P10 → B11
B11	5 V accessory strip	(+205, -105, +5)	P11 up G1 → amp T8
B12	Pololu D24V90F5	(+140, -20, +8)	P12 reserved; output taped

UPPER DECK (z ref = +120) · MAST · PASS-THROUGHS			
T1	RPLidar A1	(0, 0, +170)	rotor axis at origin; scan z ≈ +210
T2	Jetson Orin Nano	(+160, +65, +120)	fan up; ports -x; D1/D2/D3, P6
T3	Raspberry Pi 5	(+142, -75, +120)	active cooler; P8, D8/D9/D11/D12
T4	Powered USB hub	(+30, -95, +120)	D11 → Pi; hosts ReSpeaker
T5	ReSpeaker 4-mic	(-72, +70, +120)	flat, unobstructed; far from T7
T6	Master switch	(-90, -95, +120)	rear corner, one-motion reach
T7	Speaker 3 W 4 Ω	(+214, -78, +120)	grille fires +x (D13)
T8	MAX98357A amp	(+210, -38, +120)	D12 I²S; P11 5 V via G1
M1	OAK-D Lite (mast)	(+255, 0, +200)	12° down; 235 mm above floor
M2	Thermal + NoIR (mast)	(+255, ±45, ±195)	co-located beside M1
G1	Grommet Ø16	(+85, -60)	P6 + P8 + P11 rise
G2	Grommet Ø16	(-15, +95)	D3 Nano USB rises
G3	Grommet Ø16	(-50, -105)	P2/P3 master-switch loop

Excluded (below plate / off-robot): JGA25-371 motors + wheels, caster, GL-SFT1200 router (stays in apartment). Box inner 350 × 250 × 120; rear inner wall x = -110, front x = +240, side walls y = ±125.